

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 13



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Recap of Previous Lecture



Topic

Vertex Covering



Topic

Minimal & Minimum vertex covering



Topic

Vertex independent set



Topic

Maximal & Maximum vertex independent set



Topic

Connectivity



Topics to be Covered



Topic

Connectivity (Questions)



Topic

Euler trail , Euler Circuit and Traversable graph



Topic

Hamiltonian path, and Hamiltonian Circuit



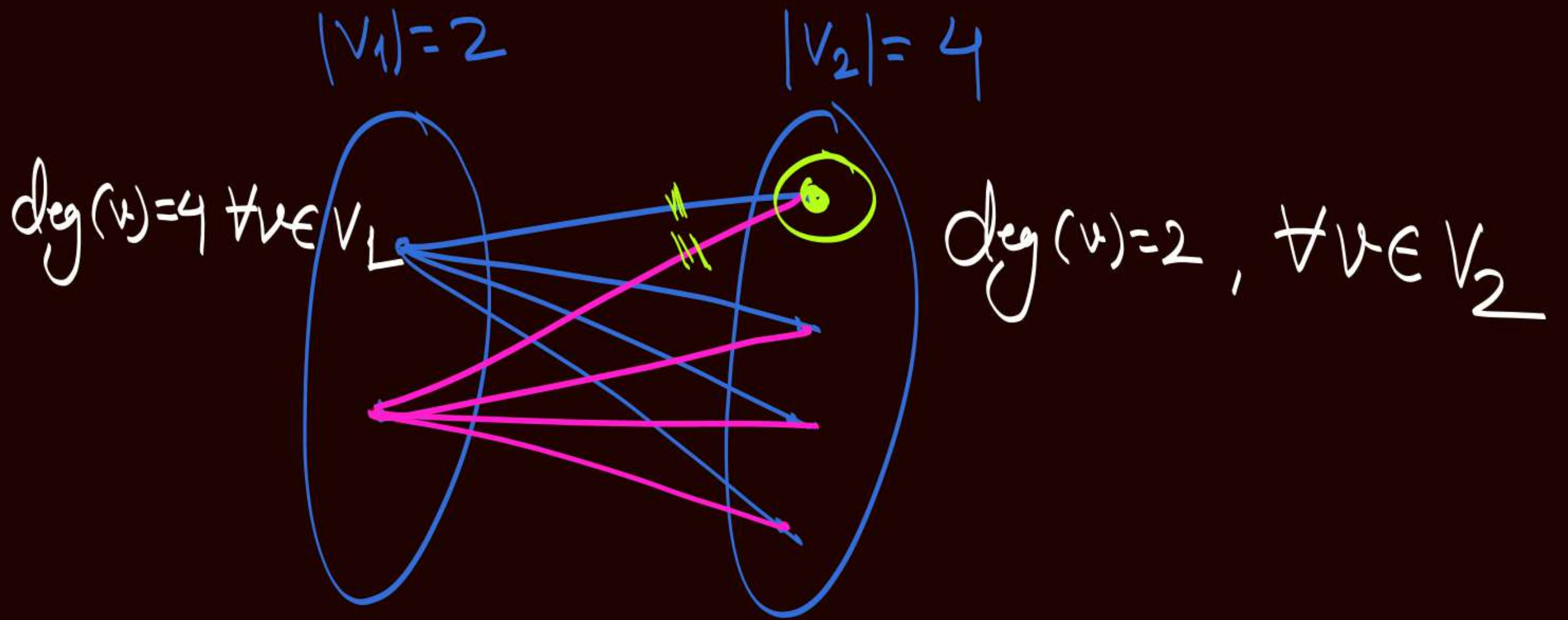
Topic

Distance, Eccentricity, Diameter, Radius and Girth



H.W. Find Vertex Connectivity ($K(G)$) and Edge Connectivity ($\lambda(G)$)
For the following graphs.

	Vertex Connectivity	Edge Connectivity
① Complete graph K_n	$(n-1)$	$(n-1)$
② Cycle graph C_n	2	2
③ Wheel graph W_n	$1 + 2 = 3$ w.r.t. hub w.r.t. Cycle	3
④ Complete bipartite graph $K_{m,n}$	$\min(m, n)$	$\min(m, n)$
⑤ Star graph with n -vertices	1	1



H.W.

Collection of trees $\hookrightarrow$ {Acyclic Components?}

#Q. If G is a forest with n vertices and k connected components, how many edges does G have?

Already discussed

Ans = " $n - k$ "

#Q. In a connected graph G if we delete an edge then no. of components are

A 0 or 1

☒ **B** 1 or 2

C 2

D ≥ 2

{ 1. When edge is not a cut edge
2. When edge is a cut edge }

[MCQ]

H.W.

with 'n' vertices

#Q. In a connected graph G if we delete a vertex then number of components are

ie, initially only '1' component

A

≤ 2

B

1 or 2

C

Lies between 1 and $n - 1$

D

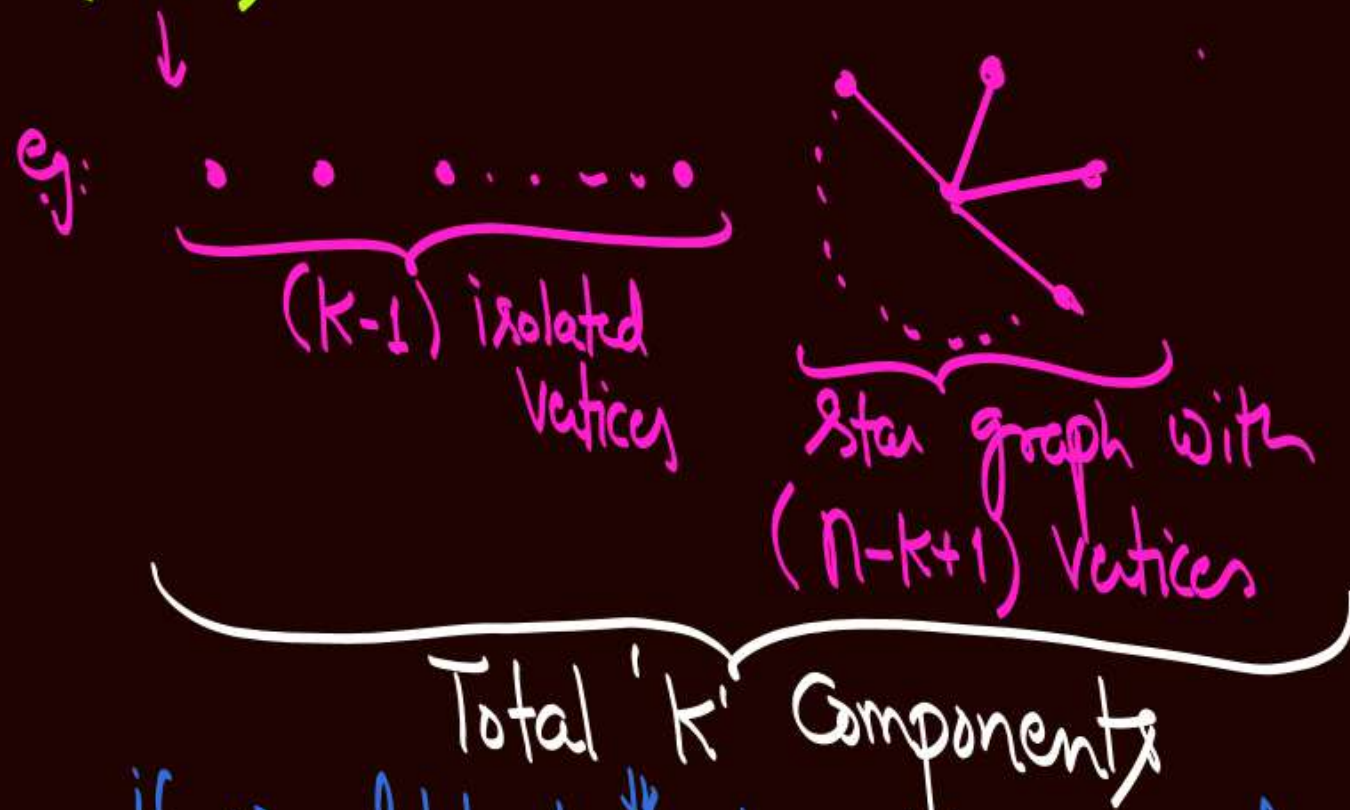
≥ 2

$\left\{ \begin{array}{l} 1, \text{ When Vertex is not a Cut vertex} \\ 'n-1', \text{ When we delete the hub vertex from star graph with 'n-vertices'} \end{array} \right\}$

Q:- Let ' G ' be a graph with ' n ' vertices
and ' k ' components, if we delete a vertex
from graph G , then number of components
will lie in the range?

Ans: $(k-1)$ to $(n-1)$

↓
When vertex that
is being deleted
is an isolated vertex



if we delete hub, then total no. of components = $(n-1)$

H.W.



#Q. A simple graph with n vertices is necessarily connected if number of edges are more than E , then find the value of E .

→ We actually need to find a value ' E ', such that if no. of edges are more than ' E ' then a simple graph can never remain disconnected

A simple graph with ' n ' vertices may be a connected graph with just $(n-1)$ edges, but a graph with n vertices and " $n-1$ " edges is not necessarily connected


'1' isolated vertex

A complete graph
with $(n-1)$ vertices

it will have
 $(n-1)C_2$ edges

In the above graph G ,

Total no. of Vertices = $1 + (n-1) = n$

Total no. of Edges = $(n-1)C_2$, and it is a disconnected graph.

* Upto $(n-1)C_2$ edges a simple graph with ' n ' vertices may remain disconnected, if we add one more edge then isolated vertex will join with '1' vertex' of $(n-1)$ vertices of Complete graph K_{n-1} . and graph will become a connected graph.

#Q. A simple graph with n vertices is necessarily connected if number of edges are more than E , then find the value of E .

→ We actually need to find a value ' E ', such that if no. of edges are more than ' E ' then a simple graph can never remain disconnected

↳ A simple graph with ' n ' vertices is necessarily connected if number of edges are more than $(n-1)C_2$

A simple graph with ' n ' vertices may be a connected graph with just $(n-1)$ edges, but a graph with n vertices and " $n-1$ " edges is not necessarily connected

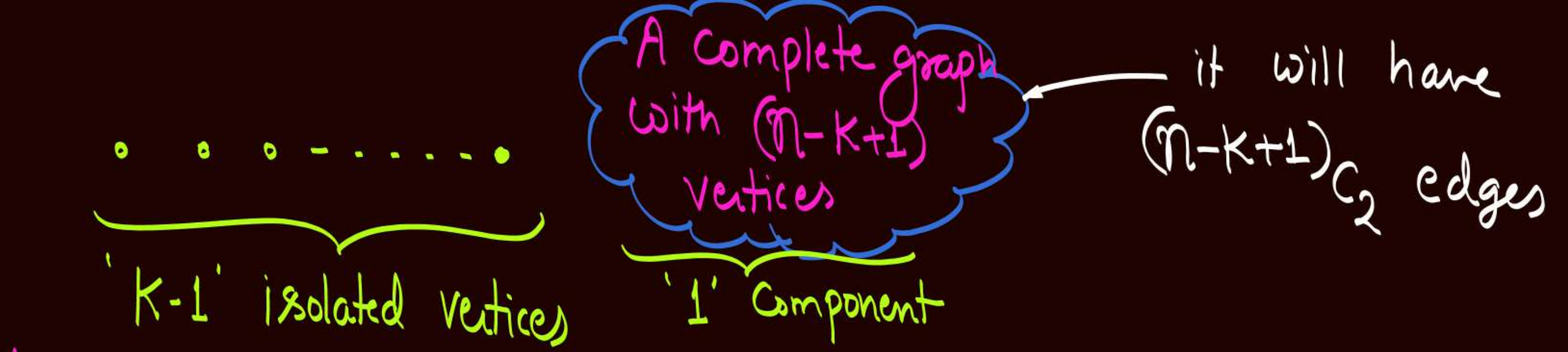
$$\therefore E = (n-1)C_2$$

H.W.



#Q. Maximum number of edges possible in ^{a simple} Δ undirected graph with n-vertices and k components?

* We are looking for number of Edges, such that if no. of edges in a simple graph with 'n' vertices are more than that value, then a simple graph with 'n' vertices can never have 'k' components.



Total no. of vertices = n

Total no. of Components = $(k-1) + 1 = k$

Total no. of Edges = $(n-k+1)C_2$

- Upto $(n-k+1)C_2$ edges it may be possible to have a simple graph with ' n ' vertices and ' k ' components, but if no. of edges are more than $(n-k+1)C_2$ then a simple graph with ' n ' vertices can never have ' k ' components

H.W.



#Q. Maximum number of edges possible in ^{a simple} Δ undirected graph with n-vertices and k components?

* We are looking for number of edges, such that if no. of edges in a simple graph with 'n' vertices are more than that value, then a simple graph with 'n' vertices can never have 'k' components.

$$\text{Maximum no. of edges possible in a simple graph with 'n' vertices and 'k' components} = \binom{n-k+1}{2} = \boxed{\frac{(n-k+1) \cdot (n-k)}{2}}$$

Note:- A simple graph with ' n ' vertices and ' k ' components will have at least ' $n-k$ ' edges and it can have at most $\frac{(n-k+1)(n-k)}{2}$ edges

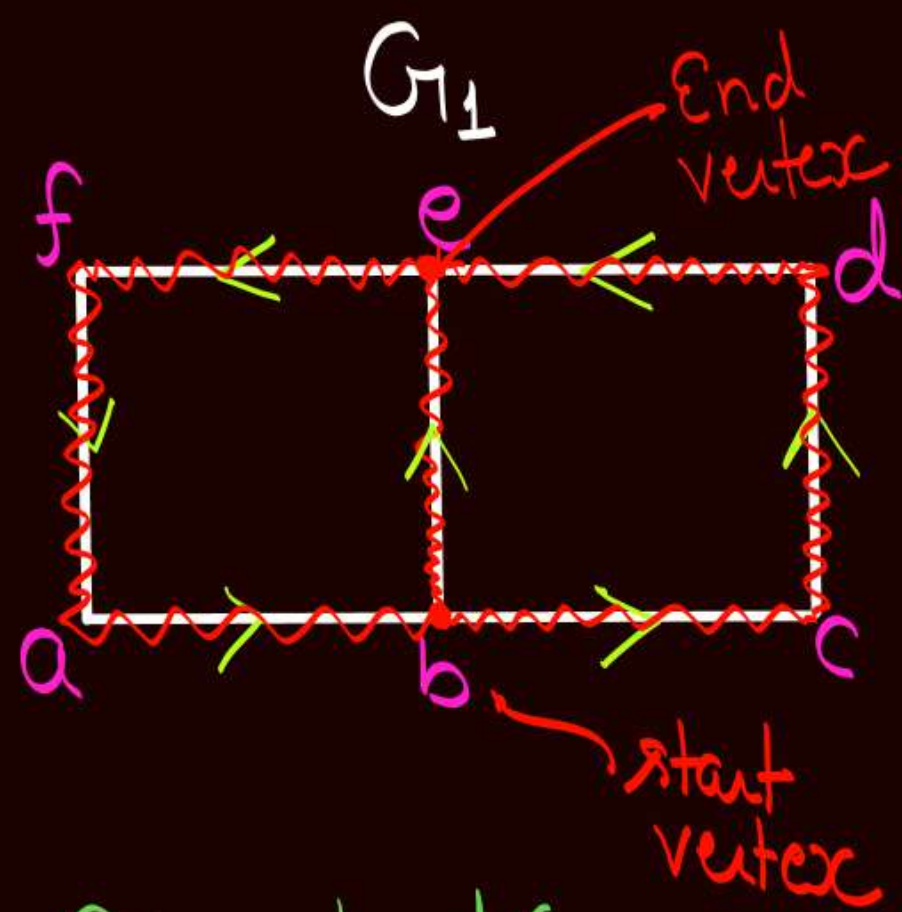


Topic : Euler trail / Euler path

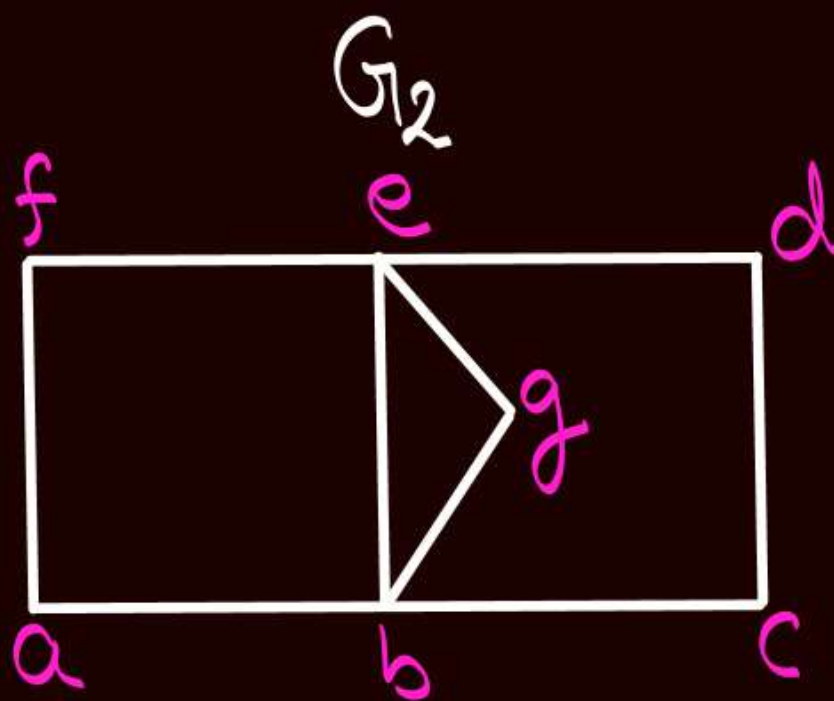
i.e., start and end vertices are different



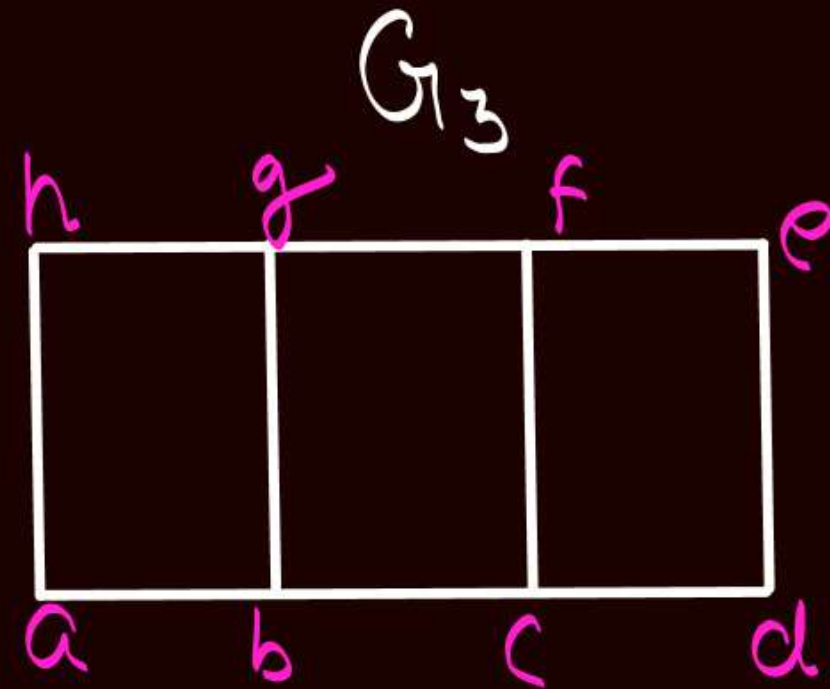
→ Let G be a Connected graph, an open trail that visit every edge of the graph exactly once and every vertex of the graph at least once is called an Euler trail / Euler path



Euler trail (open)
exists in graph G_1



No open trails
exist in G_2



No open trail
exists in G_3



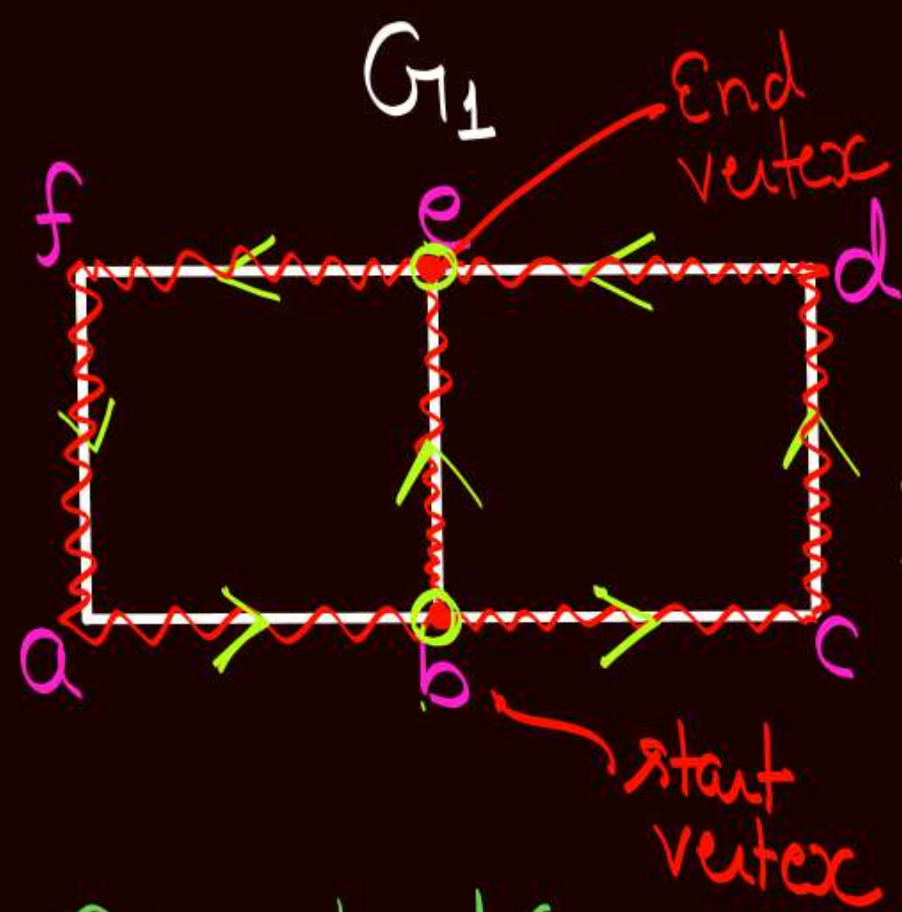
Topic : Euler circuit

i.e., closed

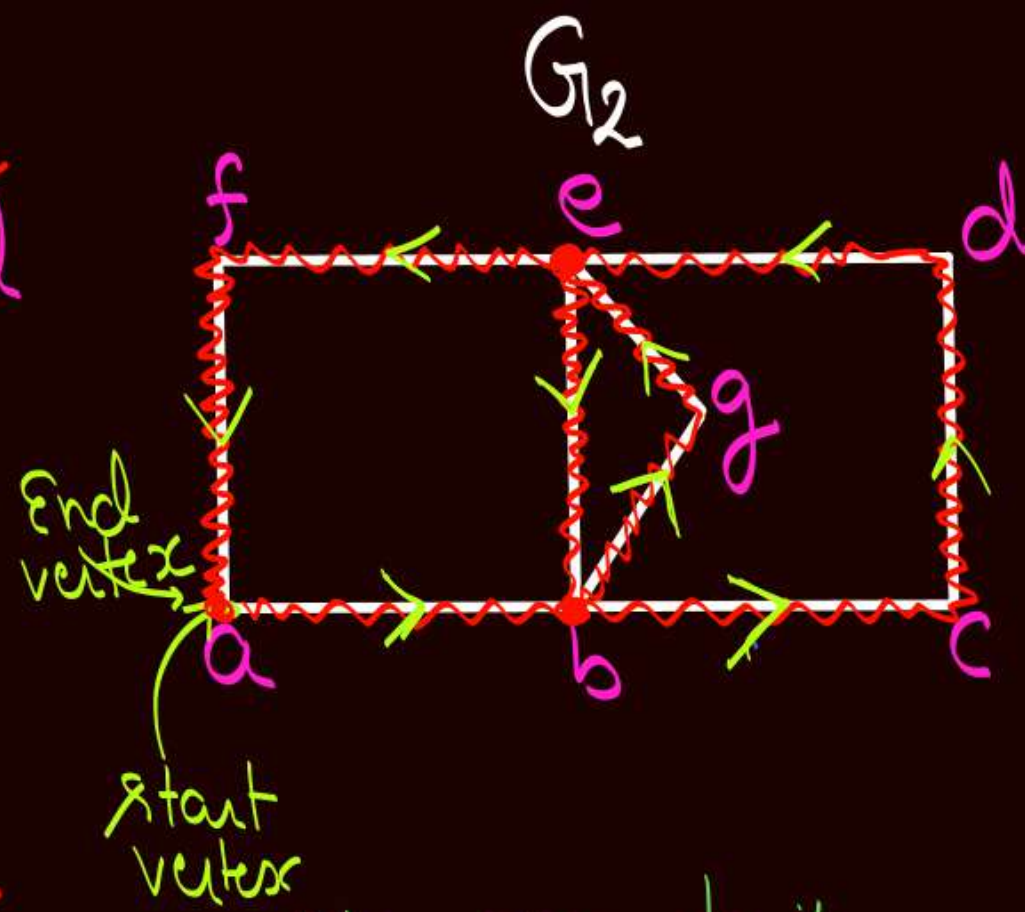
i.e., start and end vertex are same



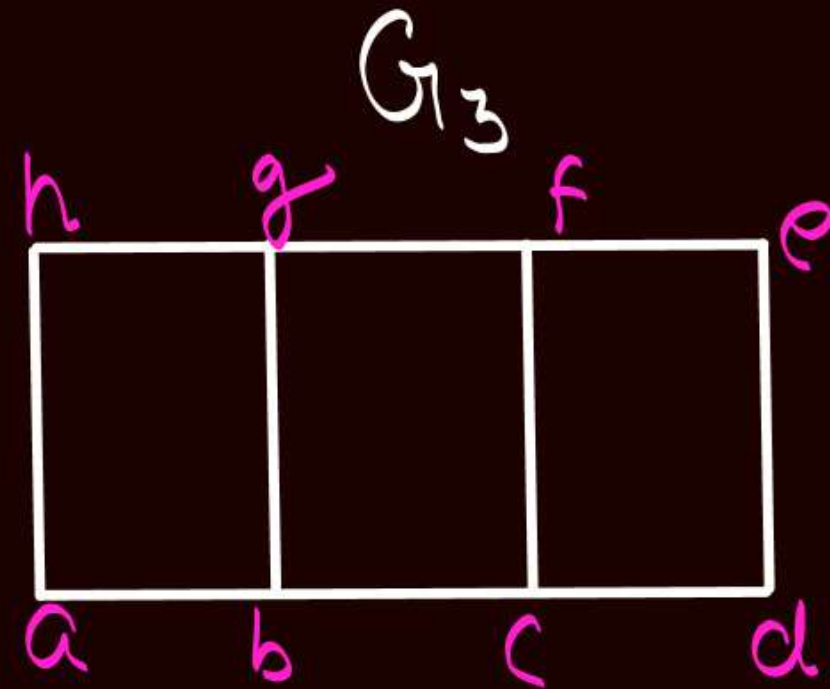
In a connected graph G , a closed trail that visit every edge of the graph exactly once and every vertex of the graph at least once, is called an "Euler Circuit"



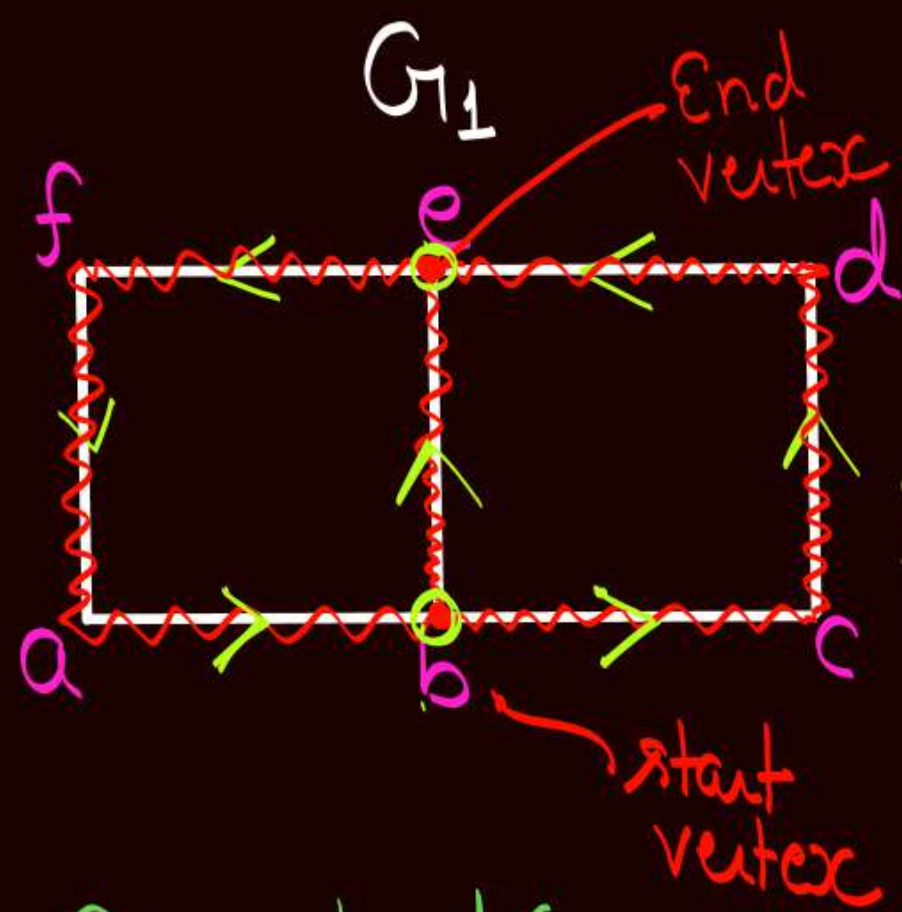
Euler trail (open)
exists in graph G_1 ,
but Euler Circuit
does not exist in G_1



No open trails
exist in G_2 ,
but Euler Circuit
exists in G_2

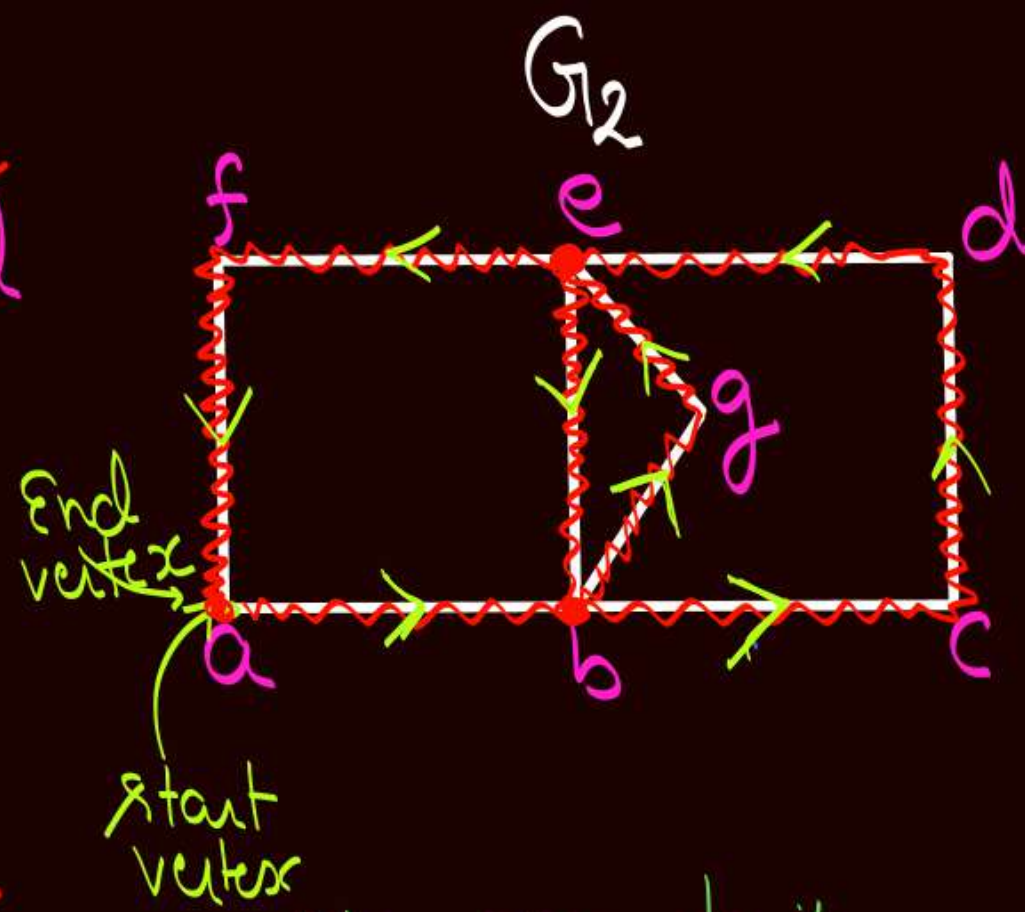


No open trail
exists in G_3
(and)
No Euler Circuit
exists in G_3



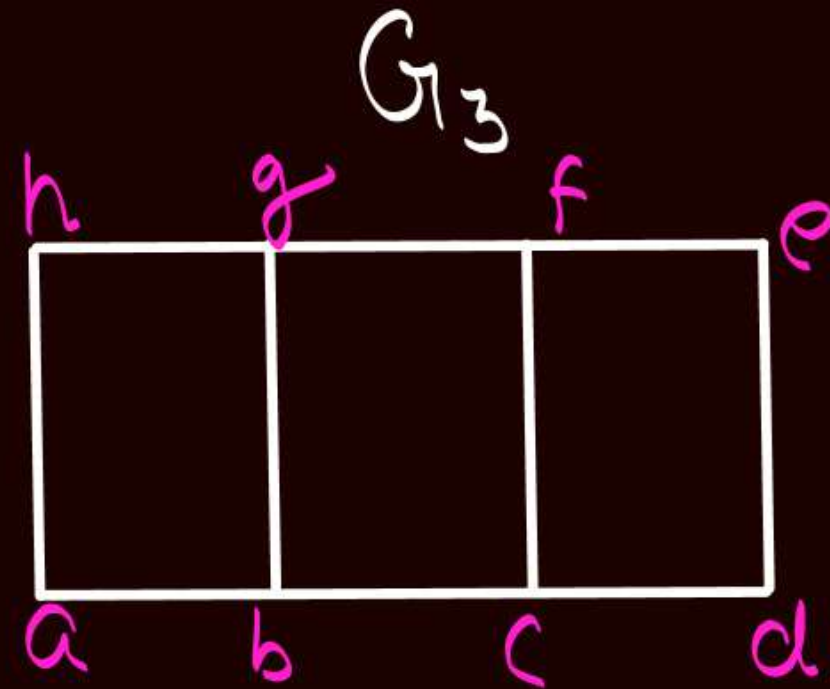
Euler trail (open)
exists in graph G_1 ,
but Euler Circuit
does not exist in G_1

→ In graph G_1
there are exactly two
vertices of odd degree



No open trails
exist in G_2 ,
but Euler Circuit
exists in G_2

→ In graph G_2
there is no vertex
of 'odd' degree



No open trail
exists in G_3
(and)

No Euler Circuit
exists in G_3

→ In G_3 more than
two vertices of
odd degree

Note:-

- ① If graph G is Connected graph, and there are no vertices of odd degree, then Euler Circuit will exist in graph G , but no Euler trail.
- ② If graph G is Connected graph, and there are exactly two vertices of odd degree, then Euler Circuit will not exist in graph G , but Euler trail exist.
- ③ If graph G is Connected graph, and there are more than two vertices of odd degree, then no Euler Circuit and no Euler trail will exist in G .

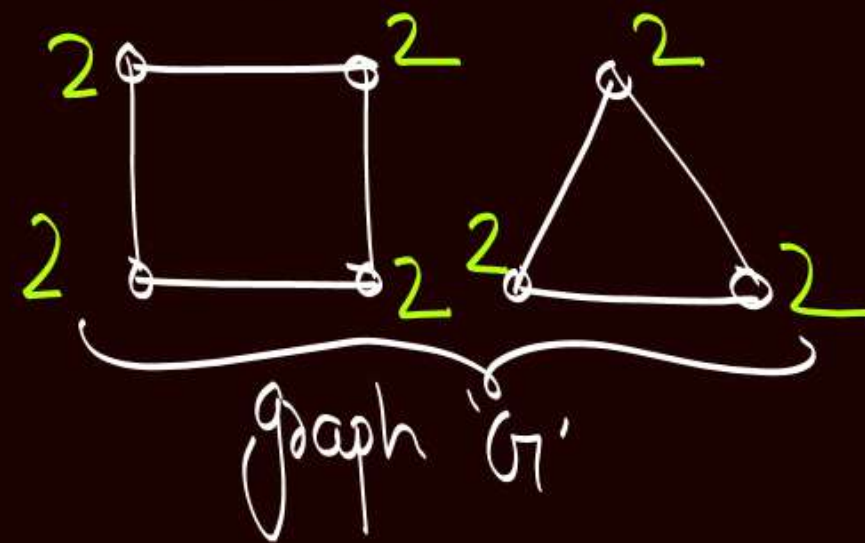
Note:- If graph G is a disconnected graph, then there will be no Euler trail and no Euler Circuit.

→ Check whether the statement is true or false,

In a graph G if all vertices are of even degree then Euler circuit exists in graph G .

→ It is a false statement, because graph may be a disconnected graph

Fig.



All vertices are of even degree, but graph is disconnected
∴ No Euler Circuit



Topic : Eulerian graph / Euler graph

➤ A graph G is an Eulerian graph if and only if Euler Circuit exists in graph G .

(or)

{ A graph G is an Eulerian graph if and only if
graph G is a Connected graph and degree of
each vertex is Even.



Topic : Traversable graph

→ A graph G is called traversable if and only if Euler circuit or Euler trail exists in graph G .

(or)

A graph G is called traversable if and only if, graph G is a connected graph and number of vertices with odd degree are '0' or '2'.



Topic : Hamiltonian path

An open path that visit
Each vertex of the graph Exactly once { we may skip? }
is called Hamiltonian Path { the edges }

start and End
vertex are
different

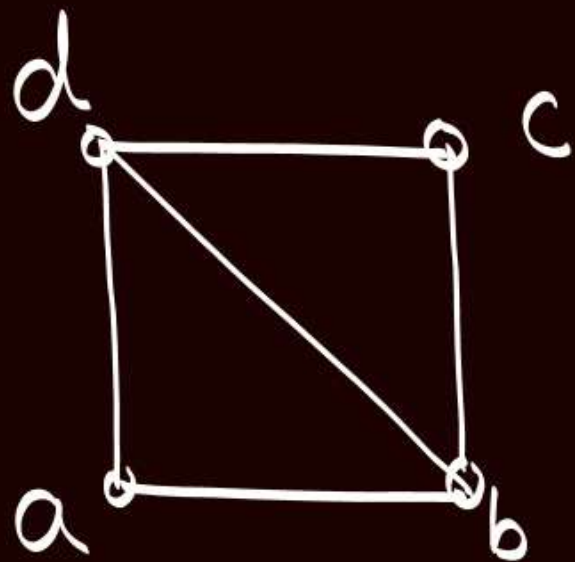
No vertex
should be
visited more
than
once



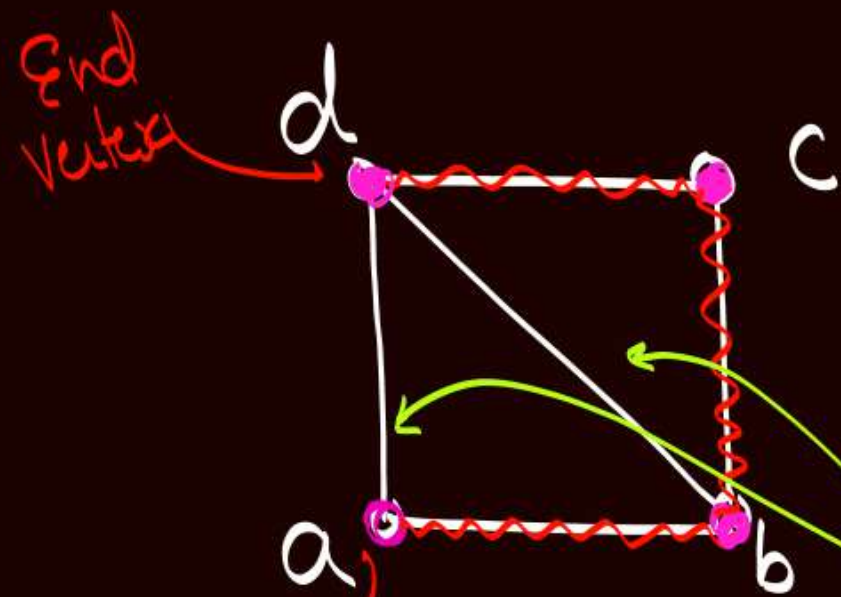
(or)

An open path that visit
Each vertex of the graph Exactly once and Each edge of the graph
at most once is called Hamiltonian path

eg



G



G

We have visited
all the vertices
exactly "once"
We have skipped
some edges

∴ Hamiltonian path
exist in graph G .



Topic : Hamiltonian circuit

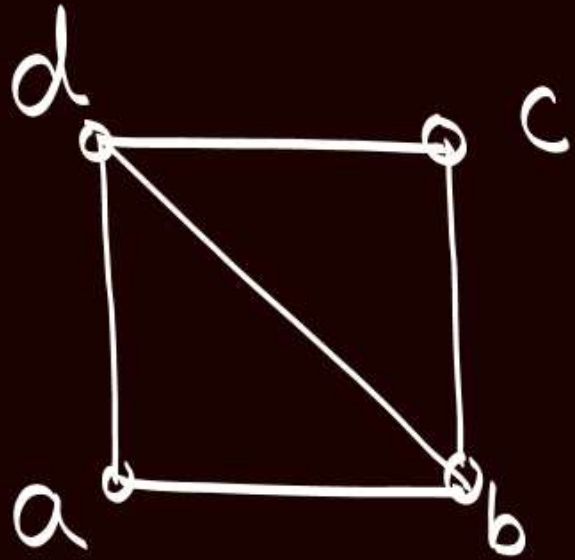


Closed

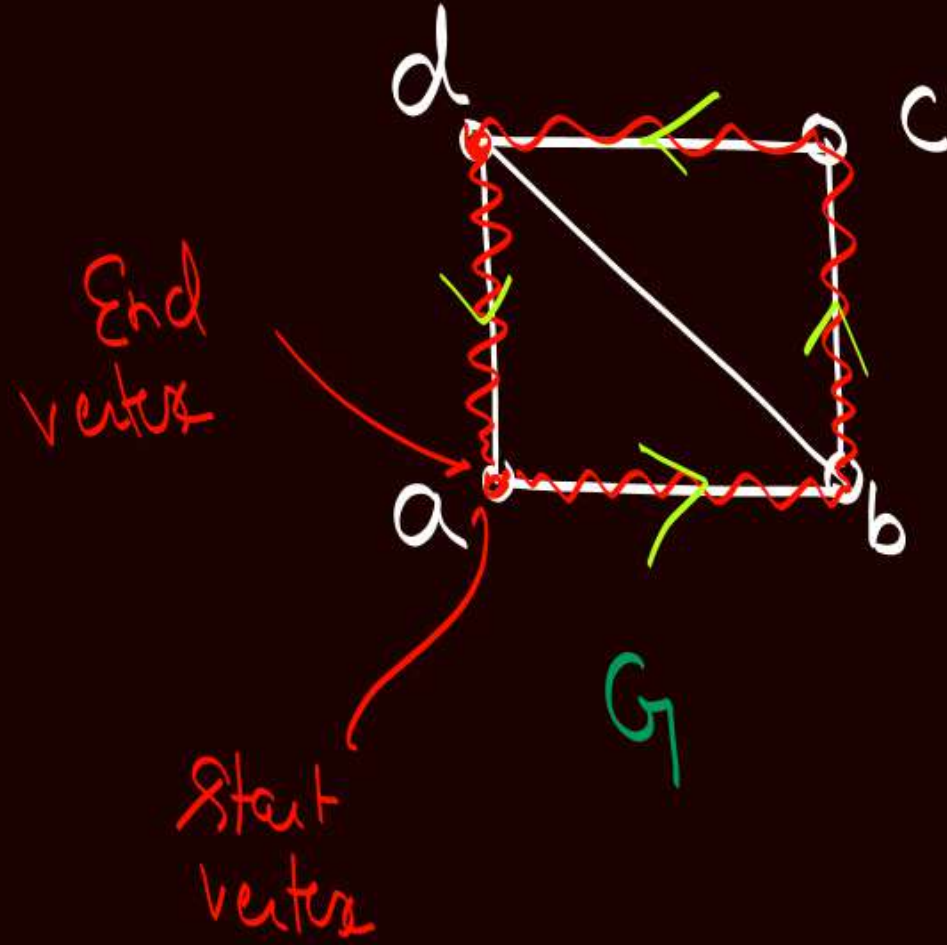
Start and end
vertex are same

A Closed Path that visit
Each vertex of the graph Exactly once Except for
the start vertex {it will be visited twice}, and each edge
of the graph at most once is called
Hamiltonian circuit.

eg



G



G

Visited Every
Vertex Exactly Once,
Visited Start vertex
Exactly twice

$\Downarrow$

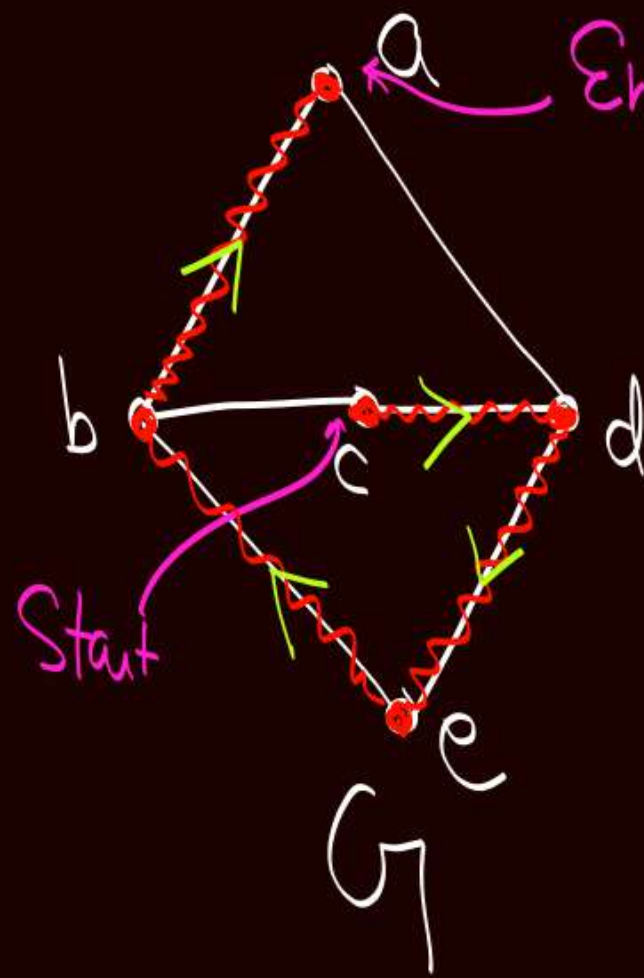
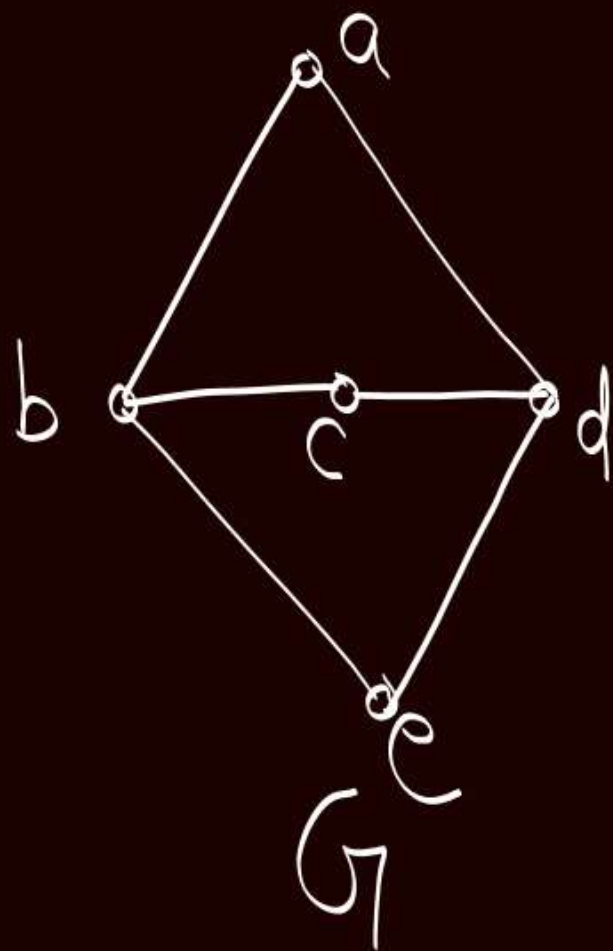
$\therefore$ Hamiltonian
Circuit exist

Note :-

If Hamiltonian Circuit exists in the graph G ,
then Hamiltonian Path will also exist in graph G .

But If Hamiltonian Path exist in graph G ,
then Hamiltonian Circuit need not exist in graph G

eg.



Hamiltonian Path
Exists in graph G ,
but no
Hamiltonian Circuit.



Topic : Hamiltonian graph

A graph G is called Hamiltonian graph
if and only if Hamiltonian circuit exist
in graph G

Note:-

There is no trick to check whether an arbitrary graph is a Hamiltonian graph or not, $\therefore$ we need to analyse the graph to answer the question.



Topic : Dirac's theorem

Let $G = (V, E)$ be a graph with n vertices in which each vertex has degree at least $n/2$, Then G has a Hamiltonian cycle.

$(n \geq 3)$

Q:- Let K_n be a complete graph where $n \geq 3$
then how many Hamiltonian cycles exist in K_n .

↳ Every Hamiltonian cycle will contain
all 'n' vertices of Complete graph K_n ,
∴ We are looking for different cycles
of length = n

- (a) 1
- (b) $(n-1)!$
- (c) $\frac{(n-1)!}{2}$
- (d) $n!$

(i) → If vertices of graph are not labeled, then Ans = 1
because all cycles of length n are identical
if vertices are not labeled

(ii) → If vertices of graph are labeled then Ans = (Home work)

Hint: Read about
Circular Permutation



2 mins Summary



✓
Topic

Connectivity (Questions)

✓
Topic

Euler trail , Euler Circuit and Traversable graph

✓
Topic

Hamiltonian path, and Hamiltonian Circuit

★
Topic

Distance, Eccentricity, Diameter, Radius and Girth

THANK - YOU